Note

If $x<0$, $0<e^{x}<1$

$x=0$, $e^{x}=1$

$x>0$; $e^{x}>1$

Indicator function condition true return 1 or else 0

$$e = \frac{\sum_{i=1}^{N} w^{(i)} \, I\{ y^{(i)} \neq f_b(x; \theta_b) \}}{\sum_{i=1}^{n} w^{(i)}}$$

$$P_k = \frac{\sum_{i \in \text{class } k} w_i}{\sum w_i}$$

$x_2 = 1.5, 2.5, 3.5$

$\boxed{x<1.5}$

Y: $\boxed{S_2}$ $+1$

N: $S_1 S_3 S_4$ -1

$\varepsilon^{(1)}$ 0.25, $\varepsilon^{(2)}$ 0.25, $\varepsilon^{(3)}$ 0.25, $\varepsilon^{(4)}$ 0 — 0.75

$\boxed{x<2.5}$

N: $S_3 S_4$ -1

Y: $S_1 S_2$ $+1$

$\varepsilon^{(1)} = 0$, $\varepsilon^{(2)} = 0.25$, $\varepsilon^{(3)} = 0.25$, $\varepsilon^{(4)} = 0$

$\boxed{x<3.5}$

N: S_3 -1

Y: $S_1 S_2 S_4$ $+1$

$\varepsilon^{(1)} = 0$, $\varepsilon^{(2)} = 0.25$, $\varepsilon^{(3)} = 0.25$, $\varepsilon^{(4)} = 0$

Adaboost classification

x_1	x_2	y	weight
1	2	+1	0.25
2	1	-1	0.25
4	4	+1	0.25
5	3	+1	0.25

① Initialization

$w^{(i)} = \frac{1}{N}$ for all samples

② for iteration $b = 1$ to B

a) fit a model $f_b(x, \theta_b)$ such that it minimize classification error, weighted by $w^{(i)}$

b) Set model weight

$$\lambda_b = \frac{1}{2} \log_2 \left(\frac{1-\epsilon}{\epsilon} \right)$$

$\log_2$

$$\epsilon = \frac{\sum_{i=1}^{N} w^{(i)} I \{ y^{(i)} \neq f_b (x; \theta_b) \}}{\sum_{i=1}^{N} w^{(i)}}$$

$$P_k = \frac{\sum_{i \in class_k} w^{(i)}}{\sum w^{(i)}}$$

c) update sample weights

$$w^{(i)} \leftarrow w^{(i)} e^{-\lambda_b} \Rightarrow \text{if prediction is correct}$$

$$w^{(i)} e^{\lambda_b} \Rightarrow \text{if prediction is wrong}$$

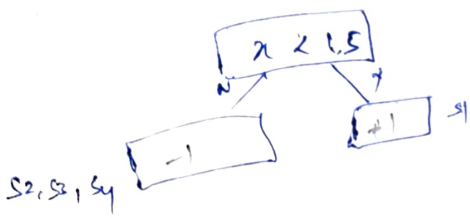
③ final model

$$F(x; \theta) = \text{sign} \left[\sum_{i=1}^{B} \lambda_b f_b (x; \theta) \right]$$

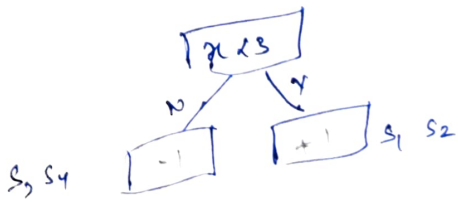
Adaboost

$$f_t = \arg\min_t \sum_i w^{(i)} I \left(y^{(i)} \neq f(x^{(i)}; \theta) \right)$$

split value of x_1 = 1.5, 3, 4.5



$\epsilon^{(1)} = 0$
$\epsilon^{(2)} = 0$
$\epsilon^{(3)} = 0.25$
$\epsilon^{(4)} = 0$
$= 0.25$ Total error



$\epsilon^{(1)} = 0$
$\epsilon^{(2)} = 0.25$
$\epsilon^{(3)} = 0.25$
$\epsilon^{(4)} = 0$
$= 0.5$

$\epsilon^{(1)} = 0$
$\epsilon^{(2)} = 0.25$
$\epsilon^{(3)} = 0$
$\epsilon^{(4)} = 0$
0.25

model weight

$$\lambda_b = \frac{1}{2} \log\left(\frac{1-\epsilon}{\epsilon}\right)$$

$$= \frac{1}{2} \log\left(\frac{1-0.25}{0.25}\right)$$

$$= \frac{1}{2} \log \frac{0.75}{0.25}$$

$$= \cancel{\frac{1}{2}}\ 0.238$$

$$\frac{1}{2} \ln\left(\frac{1-\epsilon}{\epsilon}\right)$$

$$\frac{1}{2} \ln(3)$$

$$= 0.549$$

update sample weight

$$w^{(1)} = w^{(1)} e^{-\lambda_b}$$

$$= 0.25(e^{-0.549})$$

$$= \cancel{1.44}\ 0.44$$

$$w^{(2)} = 0.25(e^{-0.549})$$

$$= \cancel{1.44}\ 0.44$$

$$w^{(3)} = 0.25(e^{0.549})$$

$$= 0.43$$

$$w^{(4)} = 0.25(e^{-0.549})$$

$$= \cancel{1.44}\ 0.144$$

x_1	x_2	y	w	Normalization
1	2	1	0.144	0.166
2	1	-1	0.144	0.166
4	4	+1	0.432	0.5
5	3	-1	0.144	0.166

Iteration 2

x_1 split value :